Artificial Intelligence — Week 1

Introduction to Artificial Intelligence

COMP3308 / COMP3608 (Russell & Norvig, ch. 1)

1 What is AI? — Four Views

AI can be characterised along two axes: *human-centred vs. rationality-centred*, and *thinking vs. acting.* This gives four possible goals for an AI system:

	Human-centred	**Rationality-centred**
Thinking	Think like humans	Think rationally
Acting	Act like humans	Act rationally

Key definitions:

- **Rationality** = ideal performance given a performance measure.
- **Acting rationally** = at each step, selecting the action that *maximises the performance measure.*

Historically all four approaches have been pursued. Sample textbook definitions map onto the views, e.g. Haugeland (1985, "machines with minds") → *think like humans*; Rich & Knight (1991) → *act like humans*; Charniak & McDermott (1985) → *think rationally*; Luger & Stubblefield (1993) and Poole et al. (1998) → *act rationally.*

2 Acting Humanly: The Turing Test Approach

Proposed by **Alan Turing (1950)**, "Computing Machinery and Intelligence". Instead of defining "thinking", replace the question *can machines think?* with a **test of behaviour**: a human interrogator exchanges written questions/answers with an unseen party; the machine passes if the interrogator cannot reliably tell whether the responses come from a human or a computer. (Inspired by the party "imitation game".)

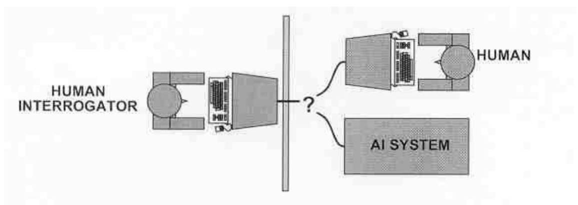

Figure 1: Turing Test setup: human interrogator on one side; a human and a computer hidden on the other; written Q&A passed between them.

Capabilities a computer needs to pass the (standard) Turing Test — the 4 major components of AI:

- **Natural language processing** — communicate in English.
- **Knowledge representation** — store what it knows/hears.
- **Automated reasoning** — answer questions, draw conclusions.

- **Machine learning** — adapt, detect and apply patterns.

Total Turing Test adds perceptual interaction (video + physical objects "through the hatch"), requiring two more disciplines:

- **Computer vision** — to perceive objects.
- **Robotics** — to manipulate/move them.

These 6 disciplines compose most of AI; the test remains relevant 70+ years on.

Historical chatbots (acting humanly):

- **ELIZA** (Weizenbaum, 1966): one of the first chatbots, mimics a psychiatrist via pattern substitution. Works on any topic but has no understanding — e.g. "my mother's poodle" → "How long has she been poodle?"
- **ALICE** (Wallace): won the Loebner Prize (restricted Turing test) three times.

Does AI try to pass the Turing Test? Mostly *no*. Simulating a human is not very useful; it is more important to understand the *principles* of intelligent behaviour and use them to build intelligent systems. *Analogy:* aeronautical engineering succeeded by learning aerodynamics, not by imitating how birds fly — the goal is not to fool pigeons.

Reverse Turing Test — CAPTCHA ("Completely Automated Public Turing test to tell Computers and Humans Apart"; von Ahn & Blum, CMU): used to block spam bots. "Reverse" because here a *machine* judges whether the other party is human.

3 Thinking Humanly: The Cognitive Modeling Approach

Goal: first build a theory of *how humans think*, then write a program that thinks the same way. Sources of insight into human thinking:

- **Introspection** — catching one's own thoughts.
- **Psychological experiments** — observing a person in action.
- **Brain imaging** — observing the brain in action.

First program in this spirit: **General Problem Solver (GPS)** (Newell & Simon, 1957) — proves theorems / solves geometric problems while imitating the *order* in which humans consider goals and sub-goals. **Cognitive science** builds and tests theories of the human mind using psychology + computer models; it is a *separate field* from AI.

4 Thinking Rationally: The "Laws of Thought" Approach

Goal: use **logic** to build intelligent systems — express a problem in logical notation and find the solution by correct inference.
Example: Socrates is a man; all men are mortal ⇒ Socrates is mortal.

Problems:

- Taking informal knowledge and formalising it is hard.
- Especially hard under *uncertainty* (knowledge <100% certain).
- Limited time/memory — which inference step to apply first?

5 Acting Rationally: The Rational Agent Approach *(the view we take)*

AI is the part of CS concerned with designing **intelligent agent programs**. An **intelligent agent**:

- has built-in knowledge and goals;
- perceives its environment;

- **acts rationally** to achieve goals, using its knowledge and the percept sequence — i.e. does the action that maximises a performance measure.

In complex environments, always doing the perfect thing is infeasible (computational demands too high), so we settle for **limited rationality**: acting appropriately when there isn't enough time for a perfect solution. (See Russell & Norvig ch. 2 for more on agents.)

6 Foundations of AI

AI draws on several older disciplines:

- **Philosophy** (from 428 B.C.): theories of reasoning, learning, rationality, logic.
- **Mathematics** (from ∼800): algorithms, formal representation and proof, computation, (un)decidability, (in)tractability, probability.
- **Psychology** (from 1879): perception, motor control, experimental methods to study the mind.
- **Computer Engineering** (from 1940): fast computers, large storage — the machines that make AI a reality.
- **Linguistics** (from 1957): structure and meaning of language, grammar, knowledge representation.

The field of **Artificial Intelligence** itself was named in **1956**.

7 A Brief History of AI

Gestation (1943–1955)

- **1943 — McCulloch & Pitts**: model of an artificial neuron — the first work recognised as AI. A neuron is binary ("on"/"off"): the weighted sum of inputs is compared to a threshold; if sum $\geq$ threshold the output is 1, else 0. Showed any computable function can be computed by a network of neurons and that AND/OR/NOT can be built from them. Parameters were hand-designed, but it was suggested they could be *learned* from examples [input, correct output].
- **1949 — Hebb**: a rule for modifying connection strengths (*Hebbian learning*), still used today.
- **1950 — Turing**: "Computing Machinery and Intelligence" — Turing test, plus a vision including machine learning, genetic algorithms and reinforcement learning; suggested *teaching* computers via learning rather than hand-programming intelligence.
- **1951 — Minsky & Edmonds**: first neural-network computer (SNARC), 40 neurons using 3000 vacuum tubes.

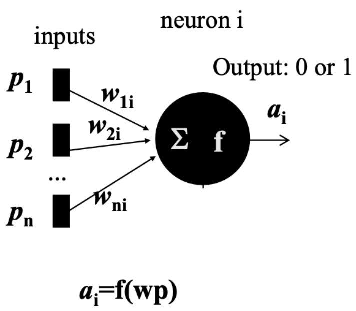

Figure 2: McCulloch–Pitts artificial neuron: inputs $p_1 \dots p_n$ with weights $w_{1i} \dots w_{ni}$, summation, threshold function f, output $a_i = f(\mathbf{w}^\top \mathbf{p})$.

Birth (1956)

John McCarthy's Dartmouth workshop coined the term "AI" — the birth of the field. A 2-month workshop, 10 participants; no breakthroughs, but the key people met. **Newell & Simon** presented the **Logic Theorist**, which proved most theorems in ch. 2 of *Principia Mathematica* (some proofs shorter than the book's). It introduced ideas central to AI: *reasoning as search in a tree* + *heuristics* to prune unpromising branches.

Early Enthusiasm (1950s–60s)

- **Perceptrons / linear neural nets** (Rosenblatt, Widrow & Hoff, ∼1960): a learning rule to train nets for pattern recognition. (Hype: the perceptron was expected to "walk, talk, see, write, reproduce itself, be conscious" — it didn't.)
- **General Problem Solver** (Newell & Simon, 1957): theorems, geometry, chess — imitating human problem solving.
- **Checkers** (Arthur Samuel, 1959–75): invented **alpha–beta pruning**; programs *learned from experience*, disproving "computers only do what they're told".
- **Lisp** (McCarthy, 1958): 2nd-oldest language still in use.
- **Resolution** (Robinson, 1965): theorem proving for first-order logic; basis for Prolog.
- **Microworld programs** (1967–68): STUDENT (Bobrow) — algebra; ANALOGY (Evans) — geometric analogies.

High Expectations

Rapid progress led to bold predictions (e.g. Herbert Simon, 1957: machines that "think, learn and create"; a chess champion and a major theorem within 10 years). But early systems *failed* when scaled to harder, broader problems.

A Dose of Reality (1960s–70s)

- **Algorithmic limits:** Minsky & Papert's book *Perceptrons* (1969) exposed limits of existing nets; more complex nets and a new learning rule were needed, but no one could yet train them.
- **Shallow knowledge:** syntactic manipulation isn't enough. Russian→English MT: "The spirit is willing but the flesh is weak" → "The vodka is good but the meat is rotten." Word sense depends on *context*; domain knowledge is required.
- **Intractability:** many real problems suffer *combinatorial explosion*; faster hardware alone doesn't solve it.

The Comeback (from the 1970s)

- **Knowledge-based / expert systems**: expert-quality, rule-based advice for narrow domains — a commercial boom. Should also provide *explanations* (e.g. a "WHY" explanation citing the rule used).
- **1980s — neural nets return**: Rumelhart & McClelland's **backpropagation** trains multilayer perceptrons, answering Minsky & Papert's criticism. (e.g. NETtalk 1987, ALVINN 1993.)

Today — Machine Learning is Reinventing Computing

ML is the fastest-growing area of AI.

- First **Deep Learning** + Big Data — multilayer nets (CNNs, LSTMs).

- Now **GenAI and LLMs** — a specialised kind of deep learning based on the **Transformer** architecture.

Model quality depends entirely on the training data ("Big Data may be Big Garbage"). LLMs/GenAI are very popular (ChatGPT hit 100M users faster than any prior technology) but carry real risks and challenges.

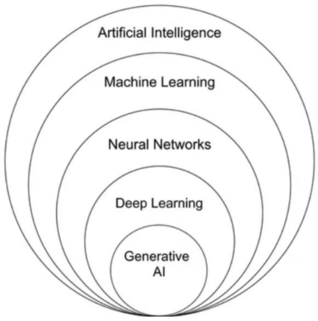

Figure 3: Artificial Intelligence ⊃ Machine Learning ⊃ Deep Learning ⊃ Generative AI / LLMs.

8 AI Today — State of the Art

Self-test (which can AI do now?): beat the world chess champion (yes); play decent table tennis; drive a curving mountain road / centre of Cairo; discover & prove a new theorem; write an intentionally funny story; give competent specialised legal advice; real-time spoken English→Mandarin; perform complex surgery. (See R&N ch. 1.4.)

Playing games — the "mice labs" of AI

Games are well-defined, test "intelligence", and have produced many cutting-edge ideas. Superhuman play now exists in chess, checkers, scrabble, backgammon, simple poker, and Go.

- **Chess — Deep Blue (IBM, 1997)** beat Kasparov (2 wins, 1 loss, 3 draws). Method: *search* (alpha–beta + iterative deepening). Chess has branching factor ≈ 35 and $\sim$50 moves/player, giving a search space on the order of 35^{100} nodes.
- **Go — AlphaGo (Google, 2016)** beat Lee Sedol 4–1. Go's 19×19 board gives branching factor $b > 300$, too large for plain search. AlphaGo uses a **convolutional neural network** (deep learning) to predict the next move, trained on $\sim$30M expert moves, backed by huge computing resources.

Other landmark applications

- **Maps & navigation**: route planning, traffic estimation, re-routing — uses AI methods, faster/better than humans.
- **Google Translate**: deep learning since 2016 (LSTMs → Transformers); trained on large parallel corpora; translates whole sentences, incl. speech and text-in-images.
- **Watson (IBM)**: won *Jeopardy!* (2011); changed public perception of AI; led to Watsonx for business (now integrates LLMs).
- **ChatGPT / LLMs (OpenAI)**: Transformer-based; trained on vast text. **GPT** = "Generative Pre-trained Transformer"; takes text as input and predicts the next text. Competitors: Gemini, Claude, Llama, Ernie, Copilot.
- **ImageNet**: $\sim$15M hand-annotated images, 20,000 categories; its recognition challenge enabled deep learning to "go big" (e.g. AlexNet, Hinton's group).

- **Self-driving cars**: Waymo (Google), Tesla, etc.; cars share collected data to learn from each other.
- **Medical diagnosis**: ML matches or beats clinicians on some image-based tasks (Alzheimer's, metastatic/breast cancer, ophthalmic, skin); focus on *human–machine partnership*.
- **Recommender systems** (Amazon, Netflix, Spotify, dating sites): method = **collaborative filtering** (unsupervised ML). Reciprocal recommenders for online dating.
- **Brain–Computer Interfaces (BCI)**: control a cursor/robotic arm using thoughts (not mind-reading) — maps EEG patterns of mental states to actions; AI task = *classification*.

Cautions about LLMs

LLMs can **hallucinate** (confident but wrong), produce eloquent yet empty answers, expose personal information, and are only "as smart as" their training data. Over-reliance harms critical thinking and problem solving. *Exciting tools, but not a replacement for humans.* (Note: ChatGPT Health, launched Jan 2026, raises sensitive privacy/safety concerns and may not match Australian clinical guidelines.)

9 Opportunities, Future Trends & the Workforce

Two modes (from *AI and Life in 2030*, Stanford):

- **AI + humans collaboratively**: e.g. analyse clinical data to suggest treatments/side-effects while a clinician chooses; early-warning systems for shock/equipment failure.
- **AI on its own**: e.g. AlphaFold (protein folding); digitising paper forms; VR training tools (surgery, disaster relief).

Sectors: transportation (smarter cars, delivery drones, on-demand matching), drug discovery, energy efficiency, home robots, healthcare (personalised diagnosis, risk prediction, monitoring), elder care, education (personalisation, auto-grading, feedback, at-risk prediction), public safety/security (fraud, spam, crime prediction).

AI & the Future of Work (US National Academies report, 2024, chaired by Tom Mitchell): profound change is coming, but roles for humans remain. AI automates routine tasks and partially automates high-skill tasks, and can help judgment/problem-solving jobs (teachers, health technicians, customer service) do better. *"AI is a powerful tool, but it is still a tool to be directed by humans."*

Key findings (summary): AI is powerful and fast-developing but its future is uncertain; it is imperfect (hallucination, bias, faulty "logic"); it can raise productivity but needs new skills; labour-market effects depend on AI's pace and human response; it may both replace and create jobs; benefits may not be shared equally; it reshapes education; better measurement is needed; and it raises major questions of fairness, privacy, safety, national security and civil discourse.

10 Take-aways & Further Resources

One-line summary: AI is fundamentally about **algorithms** (not sci-fi, magic, or just ChatGPT) — an exciting, fast-moving field you can be part of.

Suggested at home: write a poem/story with ChatGPT; watch Fei-Fei Li's TED talk; *The Smartest Machine on Earth* and *The Machine that Changed the World* (4/5) documentaries; *The Imitation Game*; read *AI and Life in 2030* and the workforce report.